Subgradient Methods

- subgradient method and stepsize rules
- convergence results and proof
- optimal step size and alternating projections
- speeding up subgradient methods

Subgradient method

subgradient method is simple algorithm to minimize nondifferentiable convex function f

$$x^{(k+1)} = x^{(k)} - \alpha_k g^{(k)}$$

- $x^{(k)}$ is the kth iterate
- $g^{(k)}$ is **any** subgradient of f at $x^{(k)}$
- $\alpha_k > 0$ is the kth step size

not a descent method, so we keep track of best point so far

$$f_{\text{best}}^{(k)} = \min_{i=1,\ldots,k} f(x^{(i)})$$

Step size rules

step sizes are fixed ahead of time

- *constant step size:* $\alpha_k = \alpha$ (constant)
- *constant step length:* $\alpha_k = \gamma / \|g^{(k)}\|_2$ (so $\|x^{(k+1)} - x^{(k)}\|_2 = \gamma$)
- *square summable but not summable:* step sizes satisfy

$$\sum_{k=1}^{\infty} \alpha_k^2 < \infty, \qquad \sum_{k=1}^{\infty} \alpha_k = \infty$$

- *nonsummable diminishing:* step sizes satisfy

$$\lim_{k \to \infty} \alpha_k = 0, \qquad \sum_{k=1}^{\infty} \alpha_k = \infty$$

Assumptions

- $f^\star = \inf_x f(x) > -\infty$, with $f(x^\star) = f^\star$
- $\|g\|_2 \leq G$ for all $g \in \partial f$ (equivalent to Lipschitz condition on f)
- $\|x^{(1)} - x^\star\|_2 \leq R$

these assumptions are stronger than needed, just to simplify proofs

Convergence results

define $\bar{f} = \lim_{k\to\infty} f_{\rm best}^{(k)}$

- *constant step size:* $\bar{f} - f^\star \leq G^2\alpha/2$, $i.e.$,
  converges to $G^2\alpha/2$-suboptimal
  (converges to $f^\star$ if f differentiable, α small enough)

- *constant step length:* $\bar{f} - f^\star \leq G\gamma/2$, $i.e.$,
  converges to $G\gamma/2$-suboptimal

- *diminishing step size rule:* $\bar{f} = f^\star$, $i.e.$, **converges**

Convergence proof

key quantity: *Euclidean distance to the optimal set*, not the function value

let $x^\star$ be any minimizer of f

$$
\begin{aligned}
\|x^{(k+1)} - x^\star\|_2^2 &= \|x^{(k)} - \alpha_k g^{(k)} - x^\star\|_2^2 \\
&= \|x^{(k)} - x^\star\|_2^2 - 2\alpha_k g^{(k)T}(x^{(k)} - x^\star) + \alpha_k^2 \|g^{(k)}\|_2^2 \\
&\leq \|x^{(k)} - x^\star\|_2^2 - 2\alpha_k (f(x^{(k)}) - f^\star) + \alpha_k^2 \|g^{(k)}\|_2^2
\end{aligned}
$$

using $f^\star = f(x^\star) \geq f(x^{(k)}) + g^{(k)T}(x^\star - x^{(k)})$

$$f(x^*) - f(x^{(k)}) \geq g^{(k)T}(x^* - x^{(k)})$$

$$-2\alpha_k (f(x^*) - f(x^{(k)})) \leq -2\alpha_k (f(x^*) - f(x^{(k)}))$$

$$-2\alpha_k (f(x^k) - f(x^*)) \geq -2\alpha_k (f(x^{(k)}) - f(x^*))$$

apply recursively to get

$$\begin{aligned}\|x^{(k+1)} - x^\star\|_2^2 &\leq \|x^{(1)} - x^\star\|_2^2 - 2\sum_{i=1}^k \alpha_i(f(x^{(i)}) - f^\star) + \sum_{i=1}^k \alpha_i^2 \|g^{(i)}\|_2^2 \\ &\leq R^2 - 2\sum_{i=1}^k \alpha_i(f(x^{(i)}) - f^\star) + G^2 \sum_{i=1}^k \alpha_i^2 \qquad (1)\end{aligned}$$

now we use

$$\sum_{i=1}^k \alpha_i(f(x^{(i)}) - f^\star) \geq (f_{\text{best}}^{(k)} - f^\star)\left(\sum_{i=1}^k \alpha_i\right)$$

to get

$$f_{\text{best}}^{(k)} - f^\star \leq \frac{R^2 + G^2 \sum_{i=1}^k \alpha_i^2}{2\sum_{i=1}^k \alpha_i}.$$

Proof

From (1), $0 \leq R^2 - 2\sum_{i=1}^k \alpha_i (f(x^{(i)}) - f^*) + G^2 \sum_{i=1}^k \alpha_i^2$

$(f_{best}^{(k)} - f^*) \sum_{i=1}^k \alpha_i \leq 2\sum_{i=1}^k \alpha_i (f(x^{(i)}) - f^*) \leq R^2 + G^2 \sum \alpha_i^2$

constant step size: for $\alpha_k = \alpha$ we get

$$f_{\text{best}}^{(k)} - f^\star \leq \frac{R^2 + G^2 k \alpha^2}{2k\alpha}$$

righthand side converges to $G^2\alpha/2$ as $k \to \infty$

constant step length: for $\alpha_k = \gamma / \|g^{(k)}\|_2$ we get

$$f_{\text{best}}^{(k)} - f^\star \leq \frac{R^2 + \sum_{i=1}^k \alpha_i^2 \|g^{(i)}\|_2^2}{2\sum_{i=1}^k \alpha_i} \leq \frac{R^2 + \gamma^2 k}{2\gamma k / G},$$

righthand side converges to $G\gamma/2$ as $k \to \infty$

square summable but not summable step sizes:
suppose step sizes satisfy

$$\sum_{i=1}^{\infty} \alpha_k^2 < \infty, \qquad \sum_{k=1}^{\infty} \alpha_k = \infty$$

then

$$f_{\text{best}}^{(k)} - f^\star \leq \frac{R^2 + G^2 \sum_{i=1}^{k} \alpha_i^2}{2 \sum_{i=1}^{k} \alpha_i}$$

as $k \to \infty$, numerator converges to a finite number, denominator converges to ∞, so $f_{\text{best}}^{(k)} \to f^\star$

Proof of step size → convergence formula

Stopping criterion

- terminating when $\dfrac{R^2 + G^2 \sum_{i=1}^k \alpha_i^2}{2 \sum_{i=1}^k \alpha_i} \leq \epsilon$ is really, really, slow

- optimal choice of α_i to achieve $\dfrac{R^2 + G^2 \sum_{i=1}^k \alpha_i^2}{2 \sum_{i=1}^k \alpha_i} \leq \epsilon$ for smallest k:

$$\alpha_i = (R/G)/\sqrt{k}, \quad i = 1, \ldots, k$$

  number of steps required: $k = (RG/\epsilon)^2$

- the truth: there really isn't a good stopping criterion for the subgradient method . . .

Proof of optimal step size :

Example: Piecewise linear minimization

$$\text{minimize} \quad f(x) = \max_{i=1,\ldots,m}(a_i^T x + b_i)$$

to find a subgradient of f: find index j for which

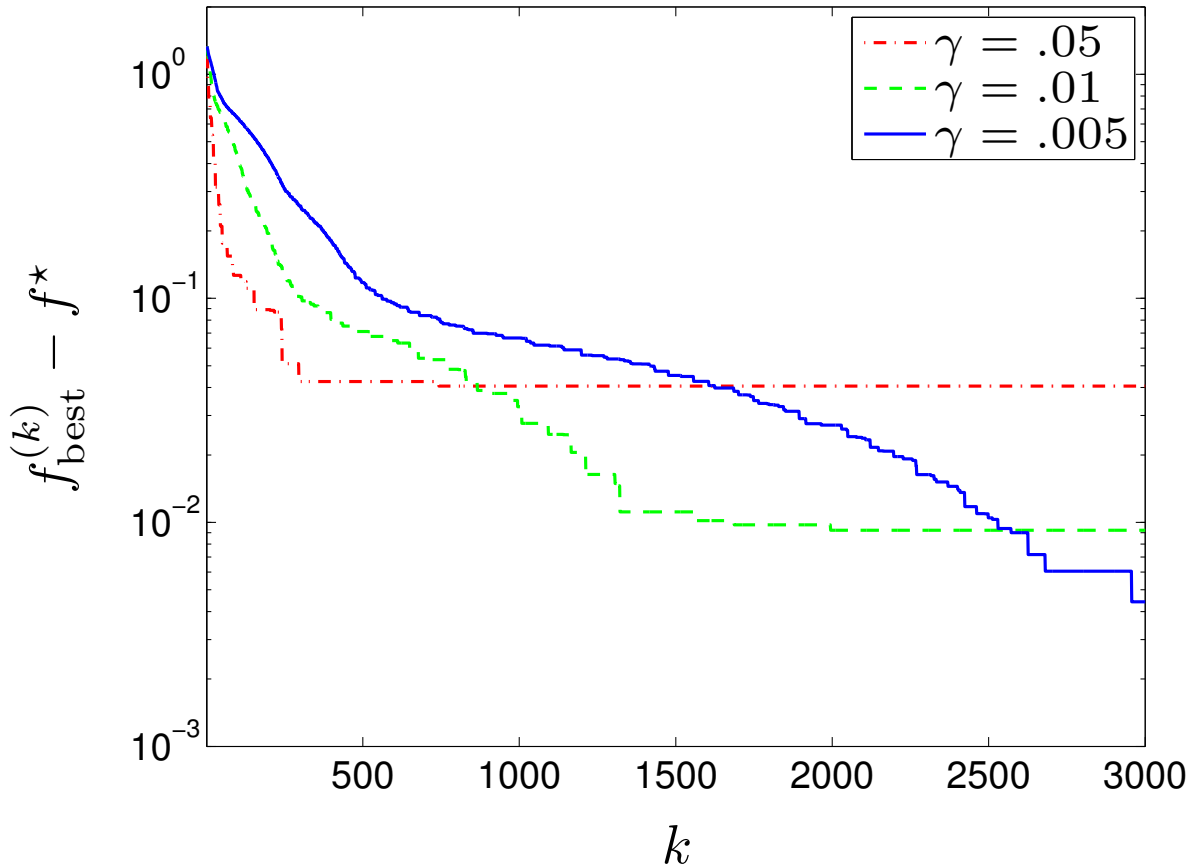
$$a_j^T x + b_j = \max_{i=1,\ldots,m}(a_i^T x + b_i)$$

and take $g = a_j$

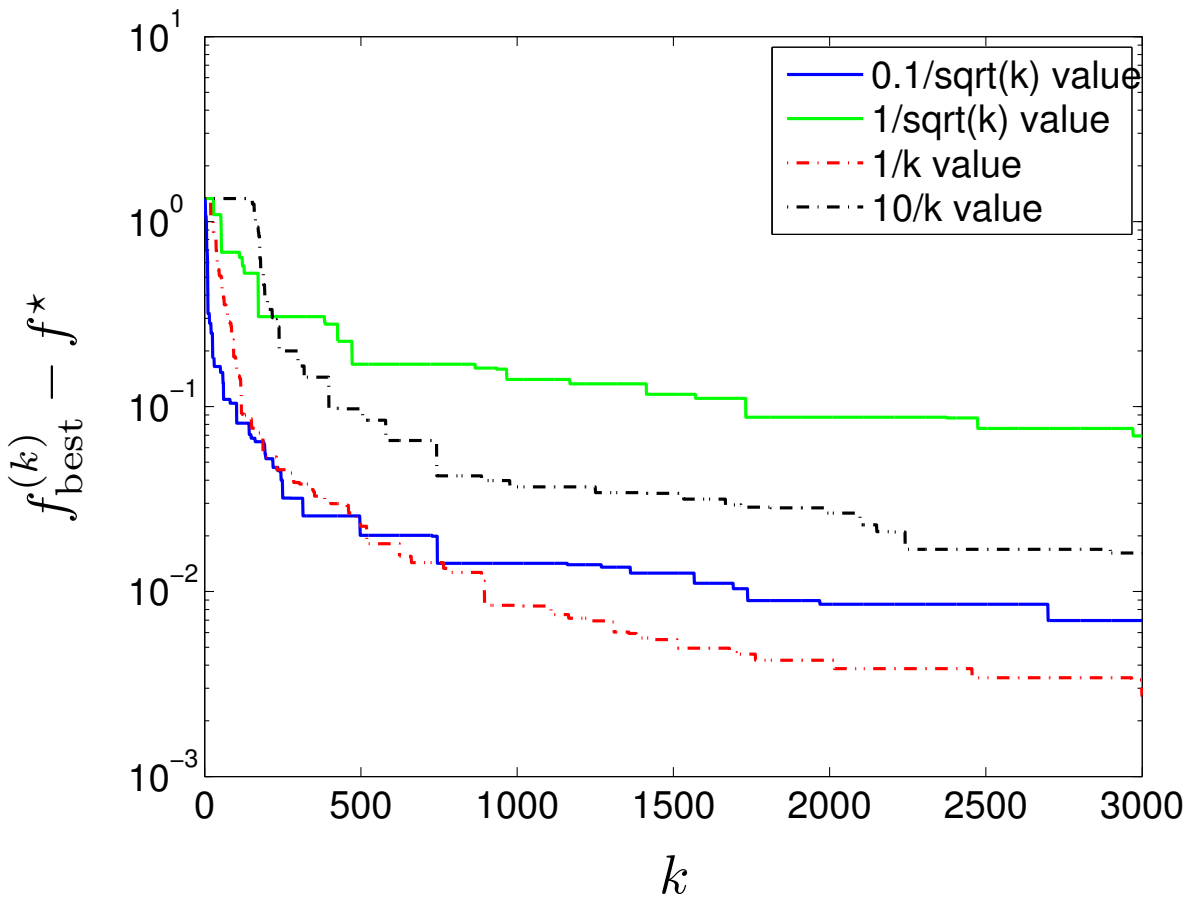
subgradient method: $x^{(k+1)} = x^{(k)} - \alpha_k a_j$

problem instance with $n = 20$ variables, $m = 100$ terms, $f^\star \approx 1.1$

$f_{\text{best}}^{(k)} - f^\star$, constant step length $\gamma = 0.05, 0.01, 0.005$



diminishing step rules $\alpha_k = 0.1/\sqrt{k}$ and $\alpha_k = 1/\sqrt{k}$, square summable step size rules $\alpha_k = 1/k$ and $\alpha_k = 10/k$



Optimal step size when $f^\star$ is known

- choice due to Polyak:

$$\alpha_k = \frac{f(x^{(k)}) - f^\star}{\|g^{(k)}\|_2^2} \qquad (1)$$

(can also use when optimal value is estimated)

- motivation: start with basic inequality

$$\|x^{(k+1)} - x^\star\|_2^2 \leq \|x^{(k)} - x^\star\|_2^2 - 2\alpha_k(f(x^{(k)}) - f^\star) + \alpha_k^2\|g^{(k)}\|_2^2$$

and choose α_k to minimize righthand side

Set $\nabla_{\alpha_k} \text{RHS} = -2(f(x^{(k)}) - f^*)) + 2\alpha_k \|g^{(k)}\|_2^2$ to 0

$\Rightarrow$ get (1)

- yields

$$\|x^{(k+1)} - x^\star\|_2^2 \leq \|x^{(k)} - x^\star\|_2^2 - \frac{(f(x^{(k)}) - f^\star)^2}{\|g^{(k)}\|_2^2}$$

(in particular, $\|x^{(k)} - x^\star\|_2$ decreases each step)

- applying recursively,

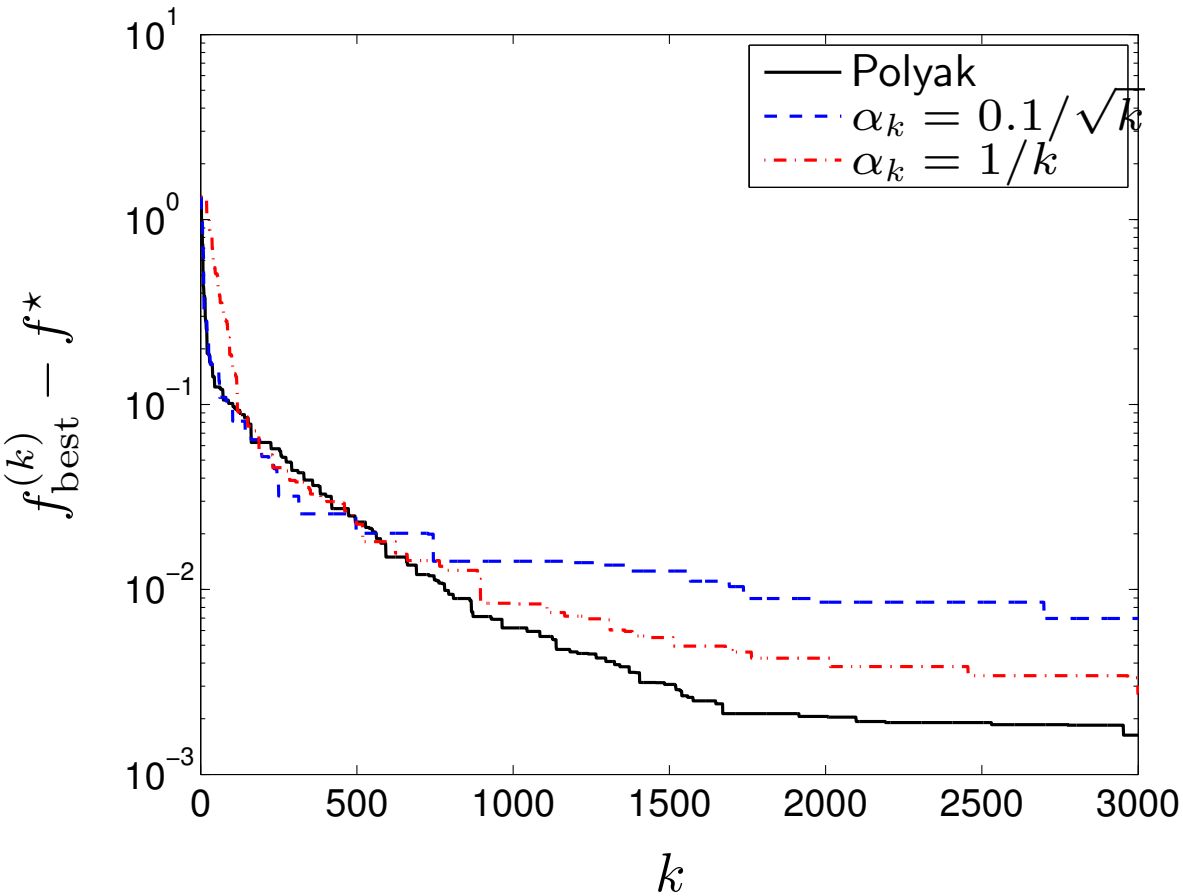
$$\sum_{i=1}^{k} \frac{(f(x^{(i)}) - f^\star)^2}{\|g^{(i)}\|_2^2} \leq R^2$$

and so

$$\sum_{i=1}^{k} (f(x^{(i)}) - f^\star)^2 \leq R^2 G^2$$

which proves $f(x^{(k)}) \to f^\star$

PWL example with Polyak's step size, $\alpha_k = 0.1/\sqrt{k}$, $\alpha_k = 1/k$



Polyak stepsize vs constant stepsize

Finding a point in the intersection of convex sets

via a subgradient method

$C = C_1 \cap \cdots C_m$ is nonempty, $C_1, \ldots, C_m \subseteq \mathbf{R}^n$ closed and convex

We can find a point in C by minimizing

$$f(x) = \max\{\mathbf{dist}(x, C_1), \ldots, \mathbf{dist}(x, C_m)\}$$

Because $f^* = 0$ (when $x^* \in C_1, C_2, \ldots, C_m$ so if we can find such x^* $\Rightarrow$ we have found a pt at the intersection of $C_1, \ldots C_m$

Then find a set C_j such that $f(x) = \text{dist}(x, C_j)$

with $\mathbf{dist}(x, C_j) = f(x)$, a subgradient of f is

$$g = \nabla \mathbf{dist}(x, C_j) = \frac{x - P_{C_j}(x)}{\|x - P_{C_j}(x)\|_2}$$

Proof : Subgradient of $dist(x, c_j)$

$$\alpha_k = \frac{f(x^{(k)}) - f(x^*)}{\|g\|_2^2} = \frac{f(x^{(k)})}{\|g\|_2^2} \quad \text{where} \quad g^{(k)} = \frac{x^{(k)} - P_{C_j}(x^{(k)})}{\|x^{(k)} - P_{C_j}(x^{(k)})\|_2}$$

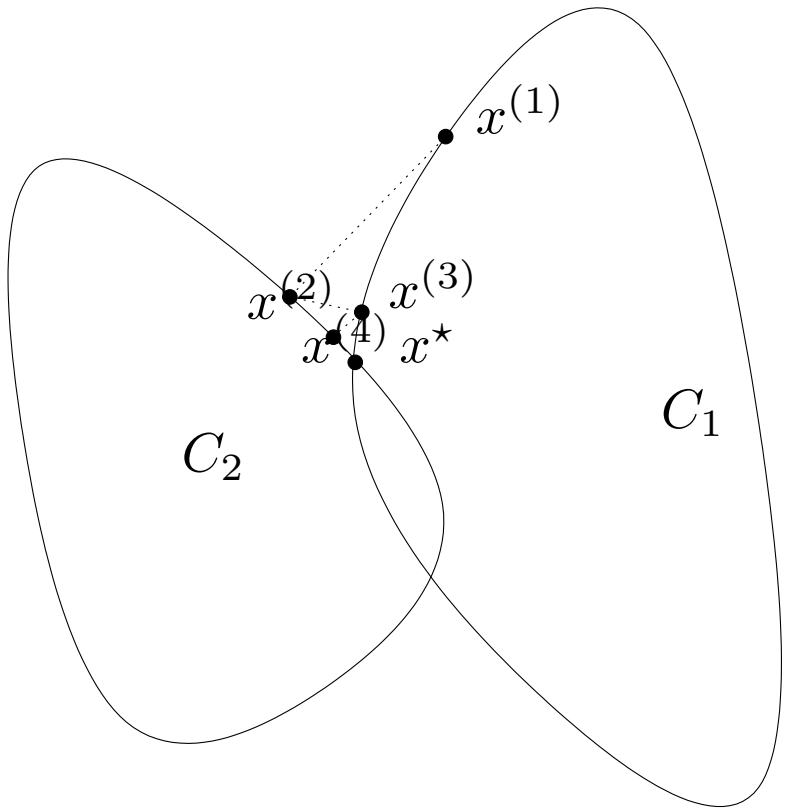
subgradient update with optimal step size:

$$\begin{aligned} x^{(k+1)} &= x^{(k)} - \alpha_k g^{(k)} \\ &= x^{(k)} - f(x^{(k)}) \frac{x^k - P_{C_j}(x^k)}{\|x^k - P_{C_j}(x^k)\|_2} \\ &= P_{C_j}(x^{(k)}) \end{aligned}$$

- a version of the famous *alternating projections* algorithm
- at each step, project the current point onto the farthest set
- for $m = 2$ sets, projections alternate onto one set, then the other
- convergence: $\mathbf{dist}(x^{(k)}, C) \to 0$ as $k \to \infty$

Alternating projections

first few iterations:



. . . $x^{(k)}$ eventually converges to a point $x^\star \in C_1 \cap C_2$

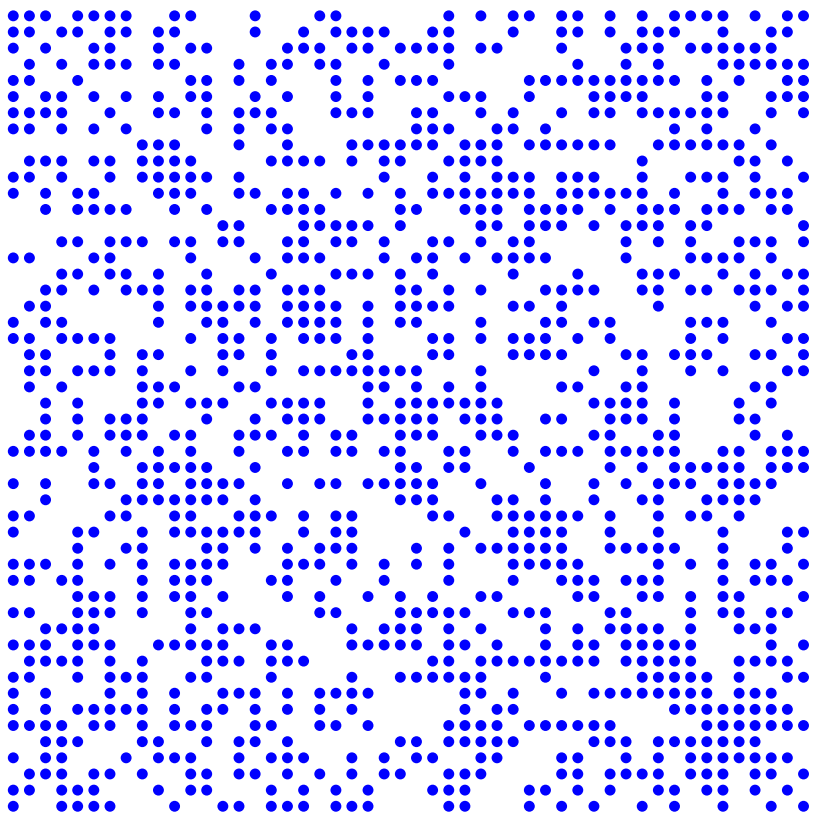
Example: Positive semidefinite matrix completion

- some entries of matrix in $\mathbf{S}^n$ fixed; find values for others so completed matrix is PSD
- $C_1 = \mathbf{S}^n_+$, C_2 is (affine) set in $\mathbf{S}^n$ with specified fixed entries
- projection onto C_1 by eigenvalue decomposition, truncation: for $X = \sum_{i=1}^n \lambda_i q_i q_i^T$,

$$P_{C_1}(X) = \sum_{i=1}^n \max\{0, \lambda_i\} q_i q_i^T$$

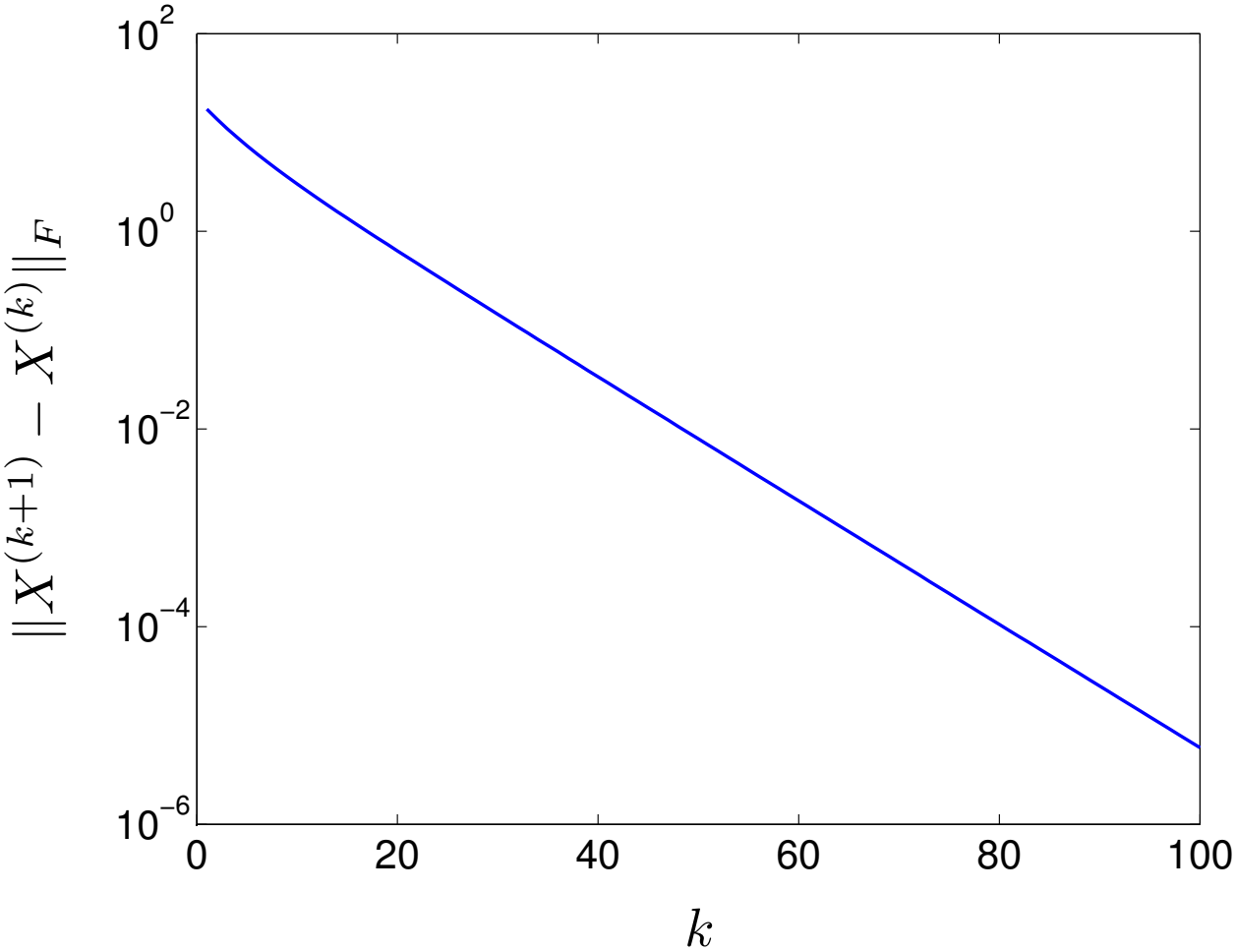
- projection of X onto C_2 by re-setting specified entries to fixed values

specific example: 50×50 matrix missing about half of its entries



- initialize $X^{(1)}$ with unknown entries set to 0

convergence is linear:



Polyak step size when $f^\star$ isn't known

- use step size

$$\alpha_k = \frac{f(x^{(k)}) - f_{\rm best}^{(k)} + \gamma_k}{\|g^{(k)}\|_2^2}$$

  with $\sum_{k=1}^\infty \gamma_k = \infty$, $\sum_{k=1}^\infty \gamma_k^2 < \infty$

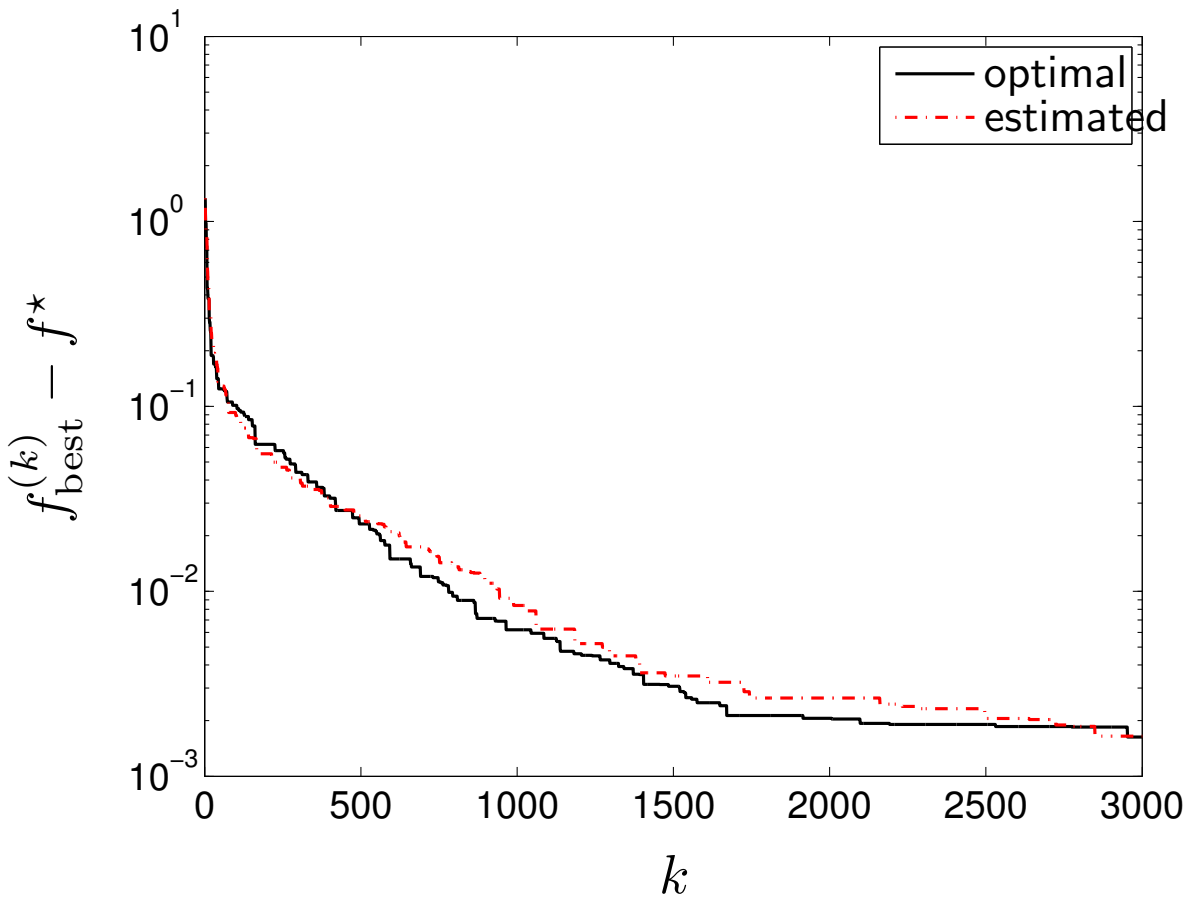
- $f_{\rm best}^{(k)} - \gamma_k$ serves as estimate of $f^\star$

- γ_k is in scale of objective value

- can show $f_{\rm best}^{(k)} \to f^\star$

Proof of convergence:

PWL example with Polyak's step size, using $f^\star$, and estimated with $\gamma_k = 10/(10+k)$



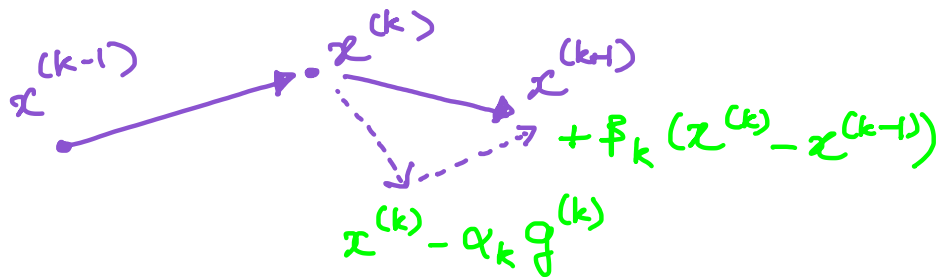
Speeding up subgradient methods

- subgradient methods are very slow
- often convergence can be improved by keeping memory of past steps

$$x^{(k+1)} = x^{(k)} - \underbrace{\alpha_k g^{(k)}}_{\text{gradient term}} + \beta_k \underbrace{(x^{(k)} - x^{(k-1)})}_{\text{Momentum term}}$$

(heavy ball method)

other ideas: localization methods, conjugate directions, . . .



A couple of speedup algorithms

$$x^{(k+1)} = x^{(k)} - \alpha_k s^{(k)}, \qquad \alpha_k = \frac{f(x^{(k)}) - f^\star}{\|s^{(k)}\|_2^2}$$

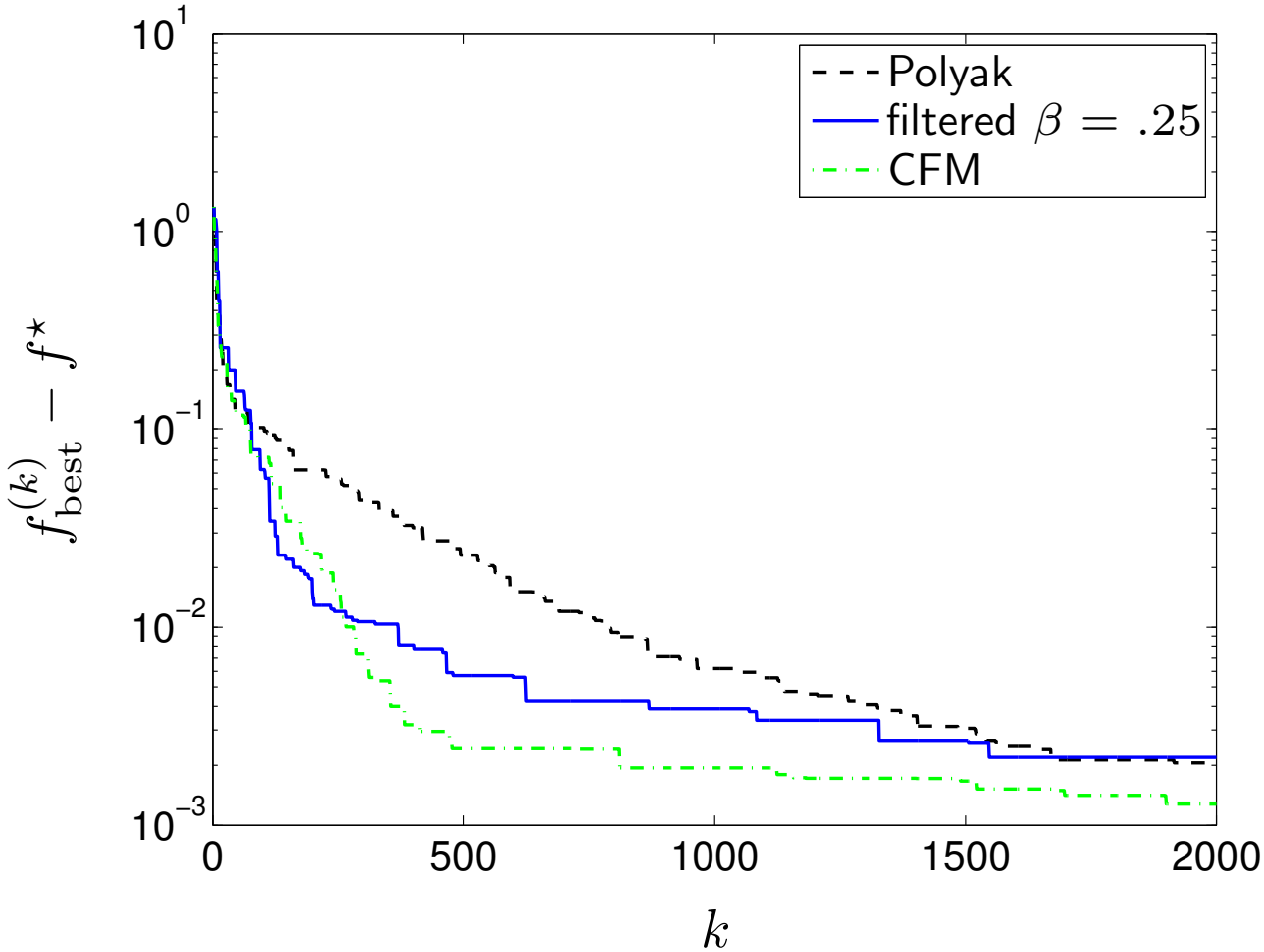
(we assume $f^\star$ is known or can be estimated)

- 'filtered' subgradient, $s^{(k)} = (1-\beta)g^{(k)} + \beta s^{(k-1)}$, where $\beta \in [0,1)$

- Camerini, Fratta, and Maffioli (1975)

$$s^{(k)} = g^{(k)} + \beta_k s^{(k-1)}, \qquad \beta_k = \max\{0, -\gamma_k (s^{(k-1)})^T g^{(k)} / \|s^{(k-1)}\|_2^2\}$$

where $\gamma_k \in [0,2)$ ($\gamma_k = 1.5$ 'recommended')

Speedup :

PWL example, Polyak's step, filtered subgradient, CFM step